

Lesson 25 — Design Challenge — Pattern Party

Matches LEGO Teacher Portal lesson: "Playground Patterns".

Objective (I can...): I can design a machine that repeats a pattern I made up, using loops. **CS concept:** Using loops for a creative build · **Builds on:** Lessons 21-24. *Also serves as flex/catch-up if the unit ran long.*

AI Connection (2 min, read aloud): Repeating is how AI practices — lots and lots of loops to get one skill right. *Ask the group:* "Would your machine get better if it practiced its pattern more, like AI does?"

Materials: LEGO CS & AI kit + device per pair (with camera), plan mats, a big screen for the coach.

Model already built (Loops model from Lesson 21) — build time is plan + code + play, small changes only.

60-minute flow *(As the flex lesson, today's clock trims Build/Code to ~18 to make room for a short AI peek. If a pair needs the full build time, run the AI peek as a whole-group demo instead.)* - **Warm-up**

(10): Loop review — pairs sort example programs into "repeat N" and "loop with many actions," and say which is simpler than copying blocks. *(Theme: Pattern party — everyone runs at once at the end.)* -

Concept + demo (8): Challenge: build a "pattern machine" (dancer, drummer, light show) using at least one loop you made. Coach says a plan with a loop, then hands off. *(Try "show a friend.")* - **Build/Code**

(~18): Pairs plan and code a repeating pattern that must use a loop on the built model. *Swap at the halfway mark.* - **Peek at a real AI — watch it recognize (10):** The **coach** opens the kit's AI pose

blocks (or Google Teachable Machine on the coach's screen) and strikes poses — arms up, arms out, hands on head — so the whole group watches the on-screen label name each pose and calls it out. **This**

is a watch-and-feed demo — the kids are exploring how a computer recognizes poses, not building their own AI. *This previews Unit 6.* **Privacy note (say it aloud):** the camera runs on the device, with no logins, and pictures are not saved. - **Share + tidy-up (10):** Pattern party — pairs run

their pattern together like a band, and each points to the loop in their program. **Photograph each build.**

Pair roles: Co-plan; Driver codes / Navigator counts the loop; swap at the halfway mark. *(Trio rotates plan/code/test.)*

Differentiation: *Support* — a single-repeat target with a partial plan. *Extension / fast-finisher* — add a second loop plus a trigger. *Reteach* — reuse a Lesson 22-24 loop as a template.

Exit check: Pair shows a working pattern that uses a loop, and can say why one repeat block is easier than copying the blocks many times (**hiding the mess**). **Progress layer:** Send home the **Unit 5**

parent note ("your child can make a machine repeat a pattern and keep count with loops") with the **build photo.** **At-home (optional):** Make up a clap pattern; say how many times each part repeats. ~5 min.